

14. Page 18: the general algorithm-summary equation

This section explains only the material shown in the page-18 image.

The displayed equation describes a very general pattern used by many smoothing algorithms:

$$p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1}) = \begin{cases} \alpha(w_i \mid w_{i-n+1}^{i-1}), & \text{if } c(w_{i-n+1}^{i-1}) > 0 \\ \gamma(w_{i-n+1}^{i-1}) p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1}), & \text{if } c(w_{i-n+1}^{i-1}) = 0 \end{cases}$$

The equation says:

To predict the next word, first check whether the current history has appeared in the training data. If it has, use the distribution for that history. If it has not, shorten the history and use a lower-order model, multiplied by a normalization factor.

The equation has two branches because the model makes two different decisions:

Current history seen?	yes → use the current-order distribution
	no → back off to a shorter history

14.1 What is the model trying to calculate?

The left-hand side is:

$$p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1})$$

Read it from inside out.

The symbol w_i means the word we are trying to predict. The expression:

$$w_{i-n+1}^{i-1}$$

means the sequence of the previous $(n - 1)$ words.

For a trigram model, $n = 3$. The history contains two previous words:

$$w_{i-n+1}^{i-1} = w_{i-2}^{i-1}$$

So the prediction becomes:

$$p_{\text{smooth}}(w_i \mid w_{i-2}, w_{i-1})$$

In plain English:

What is the probability of the next word given the previous two words?

For example:

I went to ____

If we are predicting `school`, the trigram history could be:

to school

More precisely, if the sentence fragment is:

I went to school

then, while predicting `school`, the two-word history is `went to`, and the model asks:

$$p_{\text{smooth}}(\text{school} \mid \text{went}, \text{to})$$

14.2 What does $c(w_{i-n+1}^{i-1})$ count?

The expression:

$$c(w_{i-n+1}^{i-1})$$

is the count of the current history, not the count of the complete n -gram including the predicted word.

For a trigram model, the history is two words. Therefore:

$$c(w_{i-2}, w_{i-1})$$

counts how often that two-word history occurred as a context in the training data.

Suppose the corpus contains:

```
I went to school
I went to work
They went to school
```

When predicting the word after `went to`, the trigram history is:

```
went to
```

This history occurs three times, so:

$$c(\text{went}, \text{to}) = 3$$

The complete trigram counts are different:

$$c(\text{went}, \text{to}, \text{school}) = 2$$

and:

$$c(\text{went}, \text{to}, \text{work}) = 1$$

The equation's condition checks the history count:

$$c(\text{went}, \text{to}) > 0$$

It asks whether the model has ever seen the context `went to` at all.

It does not require every possible next word to have been seen after that context.

14.3 First branch: the history has been seen

The first branch is:

$$\alpha(w_i \mid w_{i-n+1}^{i-1}) \quad \text{if } c(w_{i-n+1}^{i-1}) > 0$$

Suppose the model is predicting after the history `went to`, and that history occurred in training. Then it uses:

$$\alpha(\text{school} \mid \text{went}, \text{to})$$

The symbol α is a placeholder for the current-order smoothed distribution. The image does not define one particular formula for α ; different smoothing algorithms define it differently.

The important point is the decision:

The history exists, so stay at the current n-gram order.

For a trigram model, stay with the trigram distribution when the two-word history has been observed.

The word being predicted may still be unseen after that history. For example, `went to university` might be unseen even though `went to` is common. The displayed condition is about whether the history exists, so this branch still uses the current-order distribution α for that candidate word.

14.4 Second branch: the history has never been seen

The second branch is:

$$\gamma(w_{i-n+1}^{i-1})p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1}) \quad \text{if } c(w_{i-n+1}^{i-1}) = 0$$

This branch activates when the current history is completely unseen.

For a trigram model, the current history has two words:

$$w_{i-2}, w_{i-1}$$

If that two-word history has count zero, the model removes the oldest word and keeps only the final word:

$$p_{\text{smooth}}(w_i \mid w_{i-1})$$

So the history becomes shorter:

```
trigram context:  w_{i-2}, w_{i-1}
                  ↓ remove w_{i-2}
bigram context:   w_{i-1}
```

For example, suppose the model asks for:

$$p_{\text{smooth}}(\text{school} \mid \text{beautiful}, \text{went})$$

but the history `beautiful went` never appeared in training:

$$c(\text{beautiful}, \text{went}) = 0$$

Then the model backs off to:

$$p_{\text{smooth}}(\text{school} \mid \text{went})$$

The model is saying:

I have no evidence about the two-word context beautiful went , so I will use the shorter one-word context went .

14.5 Why is there a γ scaling factor?

When the model backs off, it does not use the lower-order distribution completely unchanged. It multiplies it by:

$$\gamma(w_{i-n+1}^{i-1})$$

The image states that γ is chosen so that the conditional distribution sums to one.

That requirement is:

$$\sum_{w_i} p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1}) = 1$$

Why might normalization be necessary? Imagine that some probability mass has already been assigned to words handled by the current-order model. The lower-order model should receive only the remaining mass, not a fresh total of 1.

A small numerical illustration:

Suppose the current-order model has already assigned probabilities to two words:

```
word A: 0.30  
word B: 0.20
```

The assigned mass is:

$$0.30 + 0.20 = 0.50$$

So the remaining mass is:

$$1 - 0.50 = 0.50$$

Suppose the lower-order model gives three backed-off words probabilities:

```
word C: 0.20
word D: 0.50
word E: 0.30
```

These already sum to one:

$$0.20 + 0.50 + 0.30 = 1.00$$

If we used them directly, the complete distribution would sum to:

$$0.30 + 0.20 + 1.00 = 1.50$$

That is impossible: probabilities must sum to one.

We therefore scale the lower-order distribution by:

$$\gamma = 0.50$$

The backed-off probabilities become:

$$P(C) = 0.50 \times 0.20 = 0.10$$

$$P(D) = 0.50 \times 0.50 = 0.25$$

$$P(E) = 0.50 \times 0.30 = 0.15$$

Now the complete distribution sums correctly:

$$0.30 + 0.20 + 0.10 + 0.25 + 0.15 = 1.00$$

The exact formula for γ depends on the smoothing algorithm, but its purpose in the displayed framework is always normalization: give the lower-order model exactly the appropriate remaining amount of probability mass.

14.6 The whole decision with a trigram example

Assume the model is predicting `school`.

Case 1: the trigram history exists

Current history:

```
went to
```

Suppose:

$$c(\text{went}, \text{to}) = 3 > 0$$

Use the first branch:

$$p_{\text{smooth}}(\text{school} \mid \text{went}, \text{to}) = \alpha(\text{school} \mid \text{went}, \text{to})$$

The model stays at the trigram level.

Case 2: the trigram history is unseen

Current history:

beautiful went

Suppose:

$$c(\text{beautiful}, \text{went}) = 0$$

Use the second branch:

$$p_{\text{smooth}}(\text{school} \mid \text{beautiful}, \text{went}) = \gamma(\text{beautiful}, \text{went}) p_{\text{smooth}}(\text{school} \mid \text{went})$$

The model backs off from a trigram context to a bigram context.

If the shorter history `went` were also unavailable in a still lower-order model, the same general idea could continue to an even shorter history. That repeated shortening is why the displayed formula is recursive.

14.7 Linear interpolation: equation (12)

The first continuation image gives the interpolation formula:

$$p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1}) = \lambda_{i-n+1}^{i-1} p_{\text{ML}}(w_i \mid w_{i-n+1}^{i-1}) + (1 - \lambda_{i-n+1}^{i-1}) p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1})$$

Let us read this slowly.

The model combines two distributions:

1. The current-order maximum-likelihood distribution:

$$p_{\text{ML}}(w_i \mid w_{i-n+1}^{i-1})$$

2. The lower-order smoothed distribution:

$$p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1})$$

The weight on the current-order distribution is:

$$\lambda_{i-n+1}^{i-1}$$

The weight on the lower-order distribution is:

$$1 - \lambda_{i-n+1}^{i-1}$$

These two weights sum to one:

$$\lambda_{i-n+1}^{i-1} + (1 - \lambda_{i-n+1}^{i-1}) = 1$$

That is why this is called **linear interpolation**: it takes a weighted linear combination of the higher-order and lower-order distributions.

A numerical example

Suppose the current history is `went to`, and we are predicting `school`. Assume:

$$\lambda_{\text{went to}} = 0.8$$

and:

$$p_{\text{ML}}(\text{school} \mid \text{went to}) = 0.50$$

The lower-order model might assign:

$$p_{\text{smooth}}(\text{school} \mid \text{to}) = 0.20$$

Then interpolation gives:

$$p_{\text{smooth}}(\text{school} \mid \text{went to}) = 0.8 \times 0.50 + (1 - 0.8) \times 0.20$$

Calculate the two contributions separately:

$$0.8 \times 0.50 = 0.40$$

$$(1 - 0.8) \times 0.20 = 0.2 \times 0.20 = 0.04$$

Therefore:

$$p_{\text{smooth}}(\text{school} \mid \text{went to}) = 0.40 + 0.04 = 0.44$$

The lower-order model contributed (0.04), even though the current history `went to` was observed. This is the defining behavior of interpolation.

14.8 Rewriting interpolation in the general α/γ form

The second continuation image says that equation (12) can be rewritten as:

$$p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1}) = \alpha'(w_i \mid w_{i-n+1}^{i-1}) + \gamma(w_{i-n+1}^{i-1})p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1})$$

where:

$$\alpha'(w_i | w_{i-n+1}^{i-1}) = \lambda_{i-n+1}^{i-1} p_{\text{ML}}(w_i | w_{i-n+1}^{i-1})$$

and:

$$\gamma(w_{i-n+1}^{i-1}) = 1 - \lambda_{i-n+1}^{i-1}$$

This is just a change of notation. The two parts of interpolation have been given shorter names:

```
 $\alpha'$  = higher-order contribution  
 $\gamma$  = lower-order weight
```

Substituting these definitions back into the rewritten formula gives the original interpolation formula:

$$\alpha'(w_i | w_{i-n+1}^{i-1}) + \gamma(w_{i-n+1}^{i-1}) p_{\text{smooth}}(w_i | w_{i-n+2}^{i-1})$$

becomes:

$$\lambda_{i-n+1}^{i-1} p_{\text{ML}}(w_i | w_{i-n+1}^{i-1}) + (1 - \lambda_{i-n+1}^{i-1}) p_{\text{smooth}}(w_i | w_{i-n+2}^{i-1})$$

So equation (12) is being placed into the same overall framework as the earlier algorithm-summary equation.

14.9 Equation (25): what is different about α ?

The image then defines:

$$\alpha(w_i | w_{i-n+1}^{i-1}) = \alpha'(w_i | w_{i-n+1}^{i-1}) + \gamma(w_{i-n+1}^{i-1}) p_{\text{smooth}}(w_i | w_{i-n+2}^{i-1})$$

This is equation (25).

The distinction between α and α' is important:

```
 $\alpha'$ : only the current-order contribution  
 $\alpha$ : the complete interpolated probability
```

In interpolation, the lower-order model contributes even when the current (n)-gram has a nonzero count. Therefore, for a seen (n)-gram, the complete current-order probability is not merely the higher-order contribution α' . It also includes the lower-order contribution weighted by γ .

Using the numerical example above:

$$\alpha'(\text{school} \mid \text{went to}) = 0.8 \times 0.50 = 0.40$$

and:

$$\gamma(\text{went to}) = 1 - 0.8 = 0.2$$

The complete α value is:

$$\alpha(\text{school} \mid \text{went to}) = 0.40 + 0.2 \times 0.20 = 0.44$$

This is exactly the interpolated probability calculated earlier.

The paper says that, by taking this definition of α , interpolated models can be placed in the general form of equation (24). It then refers to models of this form as **interpolated models**.

For the last term in equation (25), the paper says that the notation $\dots$ can be used as shorthand for the lower-order term:

$$\gamma(w_{i-n+1}^{i-1})p_{\text{smooth}}(w_i \mid w_{i-n+2}^{i-1})$$

So an expression involving $\dots$ is referring to this lower-order contribution without writing the entire term repeatedly.

14.10 The key difference between backoff and interpolated models

The image gives a precise distinction.

Backoff model

For a backoff model:

If the current n-gram has a nonzero count:
use the current-order distribution.

If the current n-gram has zero count:
use the lower-order distribution.

The lower-order distribution is not used when the current (n)-gram has a nonzero count. It is used to determine probabilities for (n)-grams with zero counts.

Interpolated model

For an interpolated model:

Use the current-order information and the lower-order information together.

The lower-order distribution contributes even when the current (n)-gram has a nonzero count. It also contributes when the current (n)-gram has a zero count.

The difference can be shown compactly:

Situation	Backoff model	Interpolated model
Current (n)-gram has nonzero count	Current-order model only	Current-order + lower-order model
Current (n)-gram has zero count	Lower-order model	Lower-order model, usually as part of the interpolation

The paper’s key sentence is therefore:

When determining the probability of an (n)-gram with a nonzero count, interpolated models use lower-order information, while backoff models do not. For (n)-grams with zero counts, both kinds of models use lower-order distributions.

This is the most important conceptual difference between the two families.

14.11 Why an interpolated model can be changed into a backoff model

The image notes that it is easy to create a backoff version of an interpolated algorithm.

In the interpolated formulation, the current-order quantity is:

$$\alpha = \alpha' + \gamma \times \text{lower-order probability}$$

To create a backoff version, use only:

$$\alpha = \alpha'$$

That means the current-order component contains only the current-order information. The lower-order model is no longer included for an observed (n)-gram.

However, after making that change, the model must adjust γ appropriately so that the complete conditional distribution still sums to one:

$$\sum_{w_i} p_{\text{smooth}}(w_i \mid w_{i-n+1}^{i-1}) = 1$$

The adjustment is necessary because probability mass has been redistributed between:

probabilities assigned at the current order
probabilities assigned after backing off

The paper closes by noting that the authors implemented both interpolated and backoff versions of several smoothing algorithms.

14.12 Complete summary of Section 2.8

The full section can be remembered as follows:

1. A general smoothing algorithm checks whether the current history has been observed.
2. Backoff models use the current-order distribution for observed histories and switch to a lower-order distribution for unseen histories.
3. Interpolated models combine current-order and lower-order distributions.
4. In interpolation, the lower-order distribution contributes even when the current (n)-gram has a nonzero count.
5. The interpolation formula can be rewritten using:

$$\alpha' = \lambda p_{\text{ML}}$$

and:

$$\gamma = 1 - \lambda$$

6. A backoff version can be formed by taking α equal to α' and adjusting γ so that probabilities still sum to one.

The shortest mental model is:

Backoff: use one order, then switch if necessary.

Interpolation: blend multiple orders all the time.